

Project 7: Expiry / Waste Risk Prediction

FMCG Inventory Intelligence — Complete Machine Learning Project

Objective: Predict which perishable FMCG products are likely to expire before being sold, so management can take preventive action and reduce waste and financial loss.

1. Problem Statement

Perishable FMCG products such as milk, yogurt, bread and juices have limited shelf lives. When demand is lower than expected, products can remain in inventory until their expiry date. This creates product waste, financial loss and inefficient inventory management. The proposed system uses historical sales, inventory and expiry information to identify products at risk before the loss occurs.

2. Real-World Example

A store has 500 yogurt cups with only 4 days remaining before expiry. If average daily sales are 70 cups, expected sales before expiry are $70 \times 4 = 280$ cups. Therefore, about 220 cups may remain unsold. The ML system should flag this inventory as high expiry/waste risk so management can act early.

Item	Example Value
Product	Yogurt
Current stock	500 cups
Remaining shelf life	4 days
Average daily sales	70 cups
Expected sales before expiry	280 cups
Potential unsold quantity	220 cups

3. Proposed Solution

Build a supervised Machine Learning classification system. Each observation represents an SKU at a particular store on a particular day. The model learns from historical examples where products became waste/expired or were sold before expiry. It then predicts the risk for current inventory.

4. Student Proposal Template — Filled

Field	Project Answer
Project Title	Expiry / Waste Risk Prediction
FMCG/Inventory Problem	Perishable products may expire before being sold, causing waste and financial loss.
Stakeholder/User	Inventory Manager, Store Manager, Supply Chain Manager
Business Decision	Decide whether to promote, discount, transfer, prioritize, or reduce replenishment for risky products.
Why ML is Required	ML can learn relationships among stock, sales velocity, shelf life, demand and promotions and provide early risk predictions.
Observation Unit	SKU $\times$ Store $\times$ Day
Target Variable (Y)	1 = likely to expire/waste before being sold; 0 = unlikely to expire/waste
Prediction Horizon	Within remaining shelf life; a next-7-day operational risk flag can also be used where appropriate.
Input Features (X)	Current stock, daily sales velocity, historical demand, remaining shelf life, days to expiry, product age, demand variability, promotion status, recent sales trend.
Dataset/Data Source	Historical sales, inventory, product expiry and promotion records from stores/warehouses. Simulated data can be used for an academic prototype.
ML Task / Algorithm	Classification using a Decision Tree; Logistic Regression or Random Forest may be used for comparison.

Field	Project Answer
Leakage Check	Use only information available at prediction time. Do not use future sales, future stock movements, or post-expiry information.
Evaluation Metrics	Precision, Recall, F1-score, Accuracy and Confusion Matrix. Recall is especially important for high-risk cases.
Decision Rules / Explainability	Short shelf life + high stock + low sales velocity → higher risk.
Prediction-to-Action	High risk → promotion/discount/transfer; Medium → monitor; Low → normal inventory handling.
Business KPI/Value	Reduced expired units, lower waste cost, improved inventory turnover and better stock utilization.
Interface/Deployment	Web dashboard showing SKU, stock, days to expiry, risk level/probability and recommended action.

5. Data Design

A suitable dataset contains one row for each SKU-store-day observation. The target records whether the product became an expiry/waste-risk case. The model uses only features that would have been known at the prediction time.

Feature	Meaning	Example
SKU_ID	Product identifier	YOG-101
Store_ID	Store identifier	S-05
Date	Observation date	2026-09-20
Stock_Qty	Current stock	500
Daily_Sales	Recent/average daily sales	70
Days_to_Expiry	Remaining days	4
Shelf_Life_Days	Total shelf life	14
Demand_Variability	Demand fluctuation	0.25
Promotion	Promotion active?	Yes
Recent_Sales_Trend	Sales direction	-0.10
Expiry_Waste_Risk	Target label	1/0

6. Machine Learning Workflow

- Collect historical inventory, sales, expiry and promotion data.
- Clean missing values, duplicate records and incorrect dates.
- Create features such as sales velocity, days-to-expiry and demand variability.
- Define the historical target label.
- Split data chronologically into training and testing sets.
- Train a Decision Tree Classifier.
- Evaluate with Precision, Recall, F1-score and Confusion Matrix.
- Predict risk for current inventory.
- Convert predictions into business actions.
- Display results in a dashboard and monitor KPIs.

7. Decision Tree Justification

A Decision Tree is suitable because it is interpretable and can represent practical business rules. For example, if days-to-expiry are low, stock is high and sales velocity is low, the tree can classify the product as high risk. This makes the prediction easier for inventory managers to understand.

8. Leakage Check

Data leakage must be prevented. Future sales, future stock transfers, actual future waste quantity, or information recorded after the prediction date must not be used as input features. A chronological train/test split is recommended because this is a time-dependent inventory problem.

9. Evaluation Metrics

Precision: Among products predicted as high risk, how many are truly high risk?

Recall: Among all truly high-risk products, how many did the model identify?

F1-score: Balance between precision and recall.

Accuracy: Overall proportion of correct predictions.

Confusion Matrix: Shows true/false positive and negative predictions.

10. Expected Decision Rules / Explainability

- High risk: short remaining shelf life + high stock + low sales velocity.
- Medium risk: moderate shelf life or uncertain demand; monitor closely.
- Low risk: sufficient shelf life + strong sales velocity; normal handling.

11. Prediction-to-Action Mapping

Risk	Recommended Action
High	Immediate discount/promotion; transfer to a high-demand store; prioritize sale; reduce or pause replenishment.
Medium	Monitor daily sales; consider targeted promotion; review upcoming purchase orders.
Low	Continue normal inventory policy and monitoring.

12. Business KPIs / Expected Value

- Reduction in expired or wasted units.
- Reduction in waste-related financial loss.
- Improvement in inventory turnover.
- Reduction in unnecessary replenishment of at-risk products.
- Better use of storage capacity.
- Improved availability of fresh products.

13. Proposed Interface / Deployment

A web-based inventory dashboard can display SKU ID, store, current stock, days to expiry, sales velocity, predicted risk, risk probability and recommended action. For a university prototype, Python with pandas and scikit-learn can be used for the model and Streamlit can be used for the interface.

14. Sample Dashboard Output

SKU	Stock	Days to Expiry	Sales/Day	Risk	Action
YOG-101	500	4	70	High	Promote / Transfer
MIL-204	120	8	35	Medium	Monitor
JUI-305	80	12	40	Low	Normal

15. Implementation Plan

Phase 1 — Data: Collect/prepare historical inventory and sales data.

Phase 2 — Preprocessing: Clean data and engineer features.

Phase 3 — Modeling: Train and validate the Decision Tree classifier.

Phase 4 — Evaluation: Compare predictions with actual outcomes.

Phase 5 — Deployment: Build the risk dashboard.

Phase 6 — Monitoring: Track waste reduction and update the model when business patterns change.

16. Final Conclusion

Expiry and waste are important inventory problems for perishable FMCG businesses. This project uses Machine Learning to identify products likely to expire before being sold. By combining stock quantity, sales velocity, remaining shelf life, demand behavior and promotion information, the system can provide an early warning. Management can then discount, promote, transfer or prioritize products before they become waste. The final goal is to reduce product waste and financial loss while improving inventory efficiency.

Complete Project Flow: Data Collection → Data Cleaning → Feature Engineering → Decision Tree Classification → Risk Prediction → Explainability → Business Action → KPI Monitoring